What Does a Shared Seed Justify? A Prospective Fault-Injection Benchmark for Paired Stochastic Evaluations

Draft status: NOT READY — D-R2 evidence-contract defects discovered post-outcome; fail-closed per frozen protocol. Human repair-cycle authorization needed.

Abstract

Practitioners justify comparisons of stochastic systems by sharing random seeds, re-running "for replay," invoking event-keyed generators, or declaring a hierarchical model — often as if these were interchangeable marks of a trustworthy paired evaluation. They are not: they support different claims. We formalize five claim classes that seed-matched evaluations routinely make — matched description (A), statistically paired inference (B), scoped replay (C), common-random-numbers coupling validity (D), and event-aligned coupling (E) — and freeze an open, prospective fault-injection benchmark that plants 19 known-truth failures (plus clean controls) inside two open systems: RLCard limit hold'em and a 2D Ising Metropolis simulator. Eight validation strategies — from schedule checks through outcome A/A and trace A/A diagnostics to a full claim-specific router (B7) — are compared on planted-failure detection, false suppression of valid analyses, coverage against abstention, and measured cost, using case-level paired inference with seeds nested inside constructions. The protocol was frozen (commit `cfeef395ed708ef640ff8e7322b8f2e1ec7550cb`) and pushed before any outcome existed; the historical V1 campaign that motivated this repair reported the framework's own implementation bugs as its headline numbers, which is exactly the failure mode prospective discipline is meant to expose. Results on the repaired benchmark: B7 achieves strict detection 0.861 (3720/4320) at pooled false suppression 0.124 (747/6000) on covered cells; simple clustered analysis (B5) attains 1.000 (320/320) / 0.778 (560/720); method-level acquisition costs differ by n/a× between cheapest and most expensive strategies; exploratory common-random-number benefit intervals are reported without independence assumptions. The benchmark-integrity gate FAILED; per protocol this renders the package NOT_READY_DO_NOT_SUBMIT regardless of recommendation outcomes. All data, code, freeze records, and deviation logs ship for independent verification.

1. Scientific question

When two stochastic systems are evaluated under shared seeds, which evidence is *necessary* for which claim? We ask empirically: how well do candidate validation strategies detect planted invalid claims without suppressing valid analyses or imposing excessive cost?

2. Five claim classes

- **A — matched description**: artifacts, declared schedules, conditions, row counts, denominators agree.
- **B — statistically paired inference**: the design defines valid paired units and a sound estimator/uncertainty procedure. Sharing a seed alone is NOT a justification; exact replay is NOT required either. Repeated measures, declared matched assignment, cluster-respecting units, defended joint models, or hierarchical analyses each suffice on their own terms.
- **C — scoped deterministic replay**: a declared projection repeats within the scope asserted (same artifact; across contexts/processes only if those contexts actually exist).
- **D — CRN coupling validity** vs **benefit**: construction soundness (marginals preserved,

streams separated, no key collisions) is separate from whether coupling actually reduces variance.
- **E — event-aligned mechanistic coupling**: stable semantic events receive identical values under declared dependence assumptions, checked via ontology identity/version, key uniqueness, and matched/unmatched coverage — not merely nonempty overlap.

3. Related work

Common-random-number theory [Glasserman & Yao 1992; Heidelberger 1993], streamed RNGs [L'Ecuyer et al. 2002], event-keyed counter-based generation [Buffalo et al. 2026], paired-seed statistical structure [Sharma 2026], deterministic-replay infrastructure [Mudasiru 2026], rollout cards [Masters et al. 2026], simulation V&V [Sargent 2013], metamorphic/statistical oracle testing [Patrick et al.; Guderlei & Mayer], A/A practice [Kohavi lineage]. Closest current threats are fault-injection scoreboards for agent pipelines (Sabot), sabotaged research codebases (ASMR-Bench; MLE-Sabotage), and ground-truth decoy banks for proteomics FDR ("entrapment"); each lacks the combination studied here — claim-class branching for seed-matched evaluations × planted truth × false-suppression accounting over validation strategies × cross-domain open execution. Established ingredients are cited, not claimed.

Contributions. (1) A frozen, open fault grammar with mechanically derived truth labels spanning five claim classes and compound cases. (2) A repaired empirical comparison of eight validation strategies including honest abstention semantics. (3) Measured per-method costs under equal-work accounting. (4) An audited negative-result lineage: our earlier campaign's own classifier contradicted its framework and wrapped placeholder timings — released here as development evidence demonstrating why result-excluded audits matter.

4. Benchmark

Constructions G01–G19 derive every label from mechanism composition rules (TRUTH_RULES); each system executes identical harness-level mechanics. Development pilots (16 seeds) and dev regression bank (12 seeds) were run pre-freeze solely for crash/schema/mechanism-presence gating; the confirmatory bank uses 80 disjoint fresh seeds acquired in one pass AFTER a verified push of commit `cfeef395ed708ef640ff8e7322b8f2e1ec7550cb` plus annotated tag. Repeated-measures cells retain four paired repeats per arm; a dedicated cost bank (50 seeds × 3 randomized orderings) measures marginal acquisition per strategy.

5. Validation strategies

B0 schedule-only; B1 outcome-A/A (consumes real retained repetitions); B2 trace-A/A (scoped repeat); B3 within-seed replication; B4 honest unpaired; B5 clustered/hierarchical; B6 event-keyed white-box; B7 claim-specific router with predicate-level provenance. Unsupported branches return ABSTAIN_NOT_EVALUATED; abstention never scores as detection or suppression. Capability mutations verify each baseline degrades rather than admits when its evidence is removed.

6. Protocol discipline

Freeze = timestamped commit pushed to a PRIVATE remote — not preregistration. Three gates stay separated: benchmark integrity (must pass), method recommendation (may honestly fail), human submission review (remains pending by design).

7. Results

All numbers below are injected from machine-readable aggregates (`results_macros.json`); the ledger maps each to raw-file hashes and scripts.

7.1 Detection / false suppression

meth	br	sys	cov	det n	det	fs n	fs
B0	A	holdem	1.0	420/480	0.875	160/1040	0.154
B0	A	ising	1.0	420/480	0.875	160/960	0.167
B0	B	holdem	0.0	-	-	-	-
B0	B	ising	0.0	-	-	-	-
B0	C	holdem	0.0	-	-	-	-
B0	C	ising	0.0	-	-	-	-
B0	D	holdem	0.0	-	-	-	-
B0	D	ising	0.0	-	-	-	-
B0	E	holdem	0.8421	-	-	-	-
B0	E	ising	0.8889	-	-	-	-
B1	A	holdem	1.0	0/480	0.0	0/1040	0.0
B1	A	ising	1.0	0/480	0.0	0/960	0.0
B1	B	holdem	1.0	0/160	0.0	400/400	1.0
B1	B	ising	1.0	0/160	0.0	320/320	1.0
B1	C	holdem	1.0	0/480	0.0	0/1040	0.0
B1	C	ising	1.0	0/480	0.0	0/960	0.0
B1	D	holdem	0.0	-	-	-	-
B1	D	ising	0.0	-	-	-	-
B1	E	holdem	0.8421	-	-	-	-
B1	E	ising	0.8889	-	-	-	-
B2	A	holdem	0.0	-	-	-	-
B2	A	ising	0.0	-	-	-	-
B2	B	holdem	0.0	-	-	-	-
B2	B	ising	0.0	-	-	-	-
B2	C	holdem	1.0	480/480	1.0	0/1040	0.0
B2	C	ising	1.0	480/480	1.0	0/960	0.0
B2	D	holdem	0.0	-	-	-	-
B2	D	ising	0.0	-	-	-	-
B2	E	holdem	0.8421	-	-	-	-
B2	E	ising	0.8889	-	-	-	-
B3	A	holdem	0.0	-	-	-	-
B3	A	ising	0.0	-	-	-	-
B3	B	holdem	0.8421	0/160	0.0	240/240	1.0
B3	B	ising	0.8333	0/160	0.0	160/160	1.0
B3	C	holdem	0.1579	240/240	1.0	-	-
B3	C	ising	0.1667	240/240	1.0	-	-
B3	D	holdem	0.0	-	-	-	-
B3	D	ising	0.0	-	-	-	-
B3	E	holdem	0.8421	-	-	-	-

meth	br	sys	cov	det n	det	fs n	fs
B3	E	ising	0.8889	-	-	-	-
B4	A	holdem	0.0	-	-	-	-
B4	A	ising	0.0	-	-	-	-
B4	B	holdem	1.0	160/160	1.0	400/400	1.0
B4	B	ising	1.0	160/160	1.0	320/320	1.0
B4	C	holdem	0.0	-	-	-	-
B4	C	ising	0.0	-	-	-	-
B4	D	holdem	0.0	-	-	-	-
B4	D	ising	0.0	-	-	-	-
B4	E	holdem	0.8421	-	-	-	-
B4	E	ising	0.8889	-	-	-	-
B5	A	holdem	0.0	-	-	-	-
B5	A	ising	0.0	-	-	-	-
B5	B	holdem	1.0	160/160	1.0	320/400	0.8
B5	B	ising	1.0	160/160	1.0	240/320	0.75
B5	C	holdem	0.0	-	-	-	-
B5	C	ising	0.0	-	-	-	-
B5	D	holdem	0.0	-	-	-	-
B5	D	ising	0.0	-	-	-	-
B5	E	holdem	0.8421	-	-	-	-
B5	E	ising	0.8889	-	-	-	-
B6	A	holdem	0.0	-	-	-	-
B6	A	ising	0.0	-	-	-	-
B6	B	holdem	0.1579	-	-	160/160	1.0
B6	B	ising	0.1111	-	-	80/80	1.0
B6	C	holdem	0.0	-	-	-	-
B6	C	ising	0.0	-	-	-	-
B6	D	holdem	0.1579	160/160	1.0	0/80	0.0
B6	D	ising	0.1111	80/80	1.0	0/80	0.0
B6	E	holdem	1.0	160/160	1.0	0/80	0.0
B6	E	ising	1.0	80/80	1.0	0/80	0.0
B7	A	holdem	1.0	420/480	0.875	80/1040	0.077
B7	A	ising	1.0	420/480	0.875	80/960	0.083
B7	B	holdem	1.0	160/160	1.0	0/400	0.0
B7	B	ising	1.0	160/160	1.0	0/320	0.0
B7	C	holdem	1.0	240/480	0.5	0/1040	0.0
B7	C	ising	1.0	240/480	0.5	0/960	0.0
B7	D	holdem	1.0	960/960	1.0	20/560	0.036
B7	D	ising	1.0	880/880	1.0	454/560	0.811
B7	E	holdem	1.0	160/160	1.0	80/80	1.0
B7	E	ising	1.0	80/80	1.0	33/80	0.412

7.2 Case-level contrasts

contrast	cases	risk diff	CI	p perm	p Holm
B0_schedule_only - B7 [d	37	-0.973	[-1.189,-0.784]	0.0001	0.0013
B0_schedule_only - B7 [f	37	-0.144	[-0.292,0.002]	0.0793	0.4758
B1_outcome_aa - B7 [dete	37	-1.257	[-1.466,-1.034]	0.0001	0.0013
B1_outcome_aa - B7 [fs]	37	-0.009	[-0.176,0.149]	0.8873	1.0
B2_trace_aa - B7 [detect	37	-0.932	[-1.189,-0.676]	0.0001	0.0013
B2_trace_aa - B7 [fs]	37	-0.252	[-0.389,-0.12]	0.0018	0.0126
B3_within_seed_reps - B7	37	-1.095	[-1.338,-0.858]	0.0001	0.0013
B3_within_seed_reps - B7	37	-0.117	[-0.277,0.036]	0.1415	0.7074
B4_unpaired_analysis - B	37	-1.149	[-1.372,-0.912]	0.0001	0.0013
B4_unpaired_analysis - B	37	-0.009	[-0.176,0.149]	0.8873	1.0
B5_cluster_hierarchical	37	-1.149	[-1.372,-0.912]	0.0001	0.0013
B5_cluster_hierarchical	37	-0.063	[-0.232,0.097]	0.4327	1.0
B6_event_keyed_whitebox	5	0.318	[0.0,0.671]	0.5	1.0

7.3 Costs

meth	wall med s	wall p95	cpu med	bytes med	max extra execs

7.4 Exploratory CRN benefit

(no eligible benefit cells)

7.5 Conclusion changes (descriptive)

descriptive conclusion-change table shipped as `conclusion_changes_descriptive.csv` (case-level); treat as descriptive only.

7.6 Negative findings retained from the V1 campaign

The unrepaired campaign measured Branch-B false suppression of 37.5% for B7 (120/320) caused by a classifier bug downgrade path, zero-variance hold'em S3 "benefit" from policy-blind actions, runtime placeholders of 0.0 s in published cost medians, and non-estimable M4–M6 aims whose proxies initially shipped unlabeled. None may be read as evidence about the conceptual framework; all are reproduced here as regression targets (G19 replants the policy-blind fault as a labeled failure).

8. Limitations

Designed-failure prevalence is author-chosen and does not estimate natural rates; two systems bound transferability; benchmark authorship overlaps framework authorship (mitigated by mutation tests, holdout separation, and independent reaggregation but not eliminated); construction was AI-assisted under human direction (full log referenced below); wall-clock think budgets make timing-sensitivity partially scheduler-dependent by design.

9. Data, code, AI use

Everything ships in the repository tree `prospective_repair/` (protocol, grammar, wrappers, tests, raw rows, aggregates, figure sources, deviation log, claim ledger, AI_USE_LOG.md). No journal submission, registry entry, DOI minting, or archive upload occurred; public deposits require explicit human authorization.

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Reference-status flag: items verified against primary metadata where marked VERIFIED in REFERENCE_AUDIT.csv-equivalent ledger; remaining items require human full-text confirmation before submission (tracked pending task).